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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 4	
TITLE : Wrapper Classes in Java	

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Programme

```
public class WrapperDemo {
    public static void main(String[] args) {
        // marks entered as Strings (as would come from user input)
        String[] marksAsString = {"85", "78", "92", "67"};

        int total = 0;

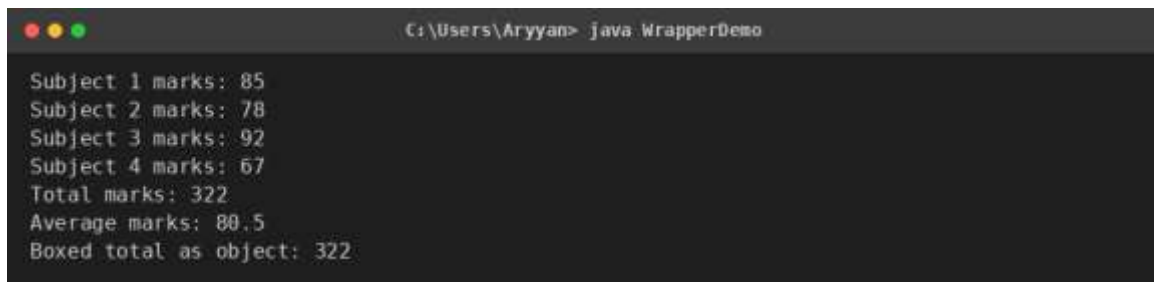
        for (int i = 0; i < marksAsString.length; i++) {
            // String -> Integer (wrapper) -> int, using parseInt
            Integer marks = Integer.parseInt(marksAsString[i]);
            total += marks;           // unboxing happens automatically here
            System.out.println("Subject " + (i + 1) + " marks: " + marks);
        }

        System.out.println("Total marks: " + total);

        double average = (double) total / marksAsString.length;
        System.out.println("Average marks: " + average);

        // demonstrating autoboxing
        Integer boxedTotal = total;    // int -> Integer, automatic
        System.out.println("Boxed total as object: " + boxedTotal);
    }
}
```

Output

A screenshot of a terminal window with a dark background. The title bar at the top shows three colored circles (red, yellow, green) on the left and the text 'C:\Users\Aryyan> java WrapperDemo' on the right. The terminal displays the following output:

```
Subject 1 marks: 85
Subject 2 marks: 78
Subject 3 marks: 92
Subject 4 marks: 67
Total marks: 322
Average marks: 80.5
Boxed total as object: 322
```

Conclusion

Thus, a Java program was developed to demonstrate the use of wrapper classes for converting between primitive data types and objects. The `Integer.parseInt()` method was used to convert String values (as would be received from user input) into usable int values, and autoboxing was used to store the result back into an Integer object. This helped in understanding how wrapper classes bridge the gap between primitive types and object-based Java features such as Collections.